ibm-granite/granite-embedding-107m-multilingual

like 52

Follow IBM Granite 5.34k

Sentence Similarity | Transformers | PyTorch | ONNX | Safetensors | 12 languages | xlm-roberta | feature-extraction | language | granite | embeddings | multilingual | mteb | Eval Results (legacy) | text-embeddings-inference | arxiv:2502.20204 | arxiv:0000.00000 | License: apache-2.0

Deploy | Copy to bucket NEW | Use this model

Model card | Files xet | Community 3

Granite-Embedding-107m-multilingual

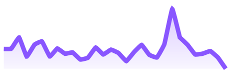
Model Summary: Granite-Embedding-107M-Multilingual is a 107M parameter dense biencoder embedding model from the Granite Embeddings suite that can be used to generate high quality text embeddings. This model produces embedding vectors of size 384 and is trained using a combination of open source relevance-pair datasets with permissive, enterprise-friendly license, and IBM collected and generated datasets. This model is developed using contrastive finetuning, knowledge distillation and model merging for improved performance.

- **Developers:** Granite Embedding Team, IBM
- **GitHub Repository:** ibm-granite/granite-embedding-models
- **Website**: Granite Docs
- **Paper:** Technical Report
- **Release Date**: December 18th, 2024
- **License:** Apache 2.0

Supported Languages: English, German, Spanish, French, Japanese, Portuguese, Arabic, Czech, Italian, Korean, Dutch, and Chinese. Users may finetune Granite-Embedding-107M-Multilingual for languages beyond these 12 languages.

Intended use: The model is designed to produce fixed length vector representations for a given text, which can be used for text similarity, retrieval, and search applications.

Downloads last month
84,799



Safetensors

Model size	0.1B pa
Tensor type	BF16

Run 15,000+ Models Instantly

Inference Providers let you run inference on thousands of models served by our partners using a simple, unified, OpenAI-compatible serverless API (Learn more).

`ibm-granite/granite-embedding-107m-multilingual` is supported by the following Inference Providers:

HF Inference API

View API Code | Dismiss

Inference Provid

Sentence Similarity

Source Sentence

Your sentence her

Sentences to compar

Your sentence here...

Your sentence here...

Add Sentence

Generate

View Code | Maximize

Model tree for `ibm-granite/grani...`

Finetunes — 12 models
Quantizations — 38 models

Spaces using `ibm-granite/gra...` 13

- mteb/leaderboard
- hakari-bench/leaderboard
- MTEB-BR/leaderboard
- droidnexus/droidnexus-labs-hub

Usage with Sentence Transformers: The model is compatible with SentenceTransformer library and is very easy to use:

First, install the sentence transformers library

```
pip install sentence_transformers
```

The model can then be used to encode pairs of text and find the similarity between their representations

```
from sentence_transformers import SentenceTra

model_path = "ibm-granite/granite-embedding-1
# Load the Sentence Transformer model
model = SentenceTransformer(model_path)

input_queries = [
    ' Who made the song My achy breaky heart'
    'summit define'
    ]

input_passages = [
    "Achy Breaky Heart is a country song wri
    "Definition of summit for English Langua
    ]

# encode queries and passages
query_embeddings = model.encode(input_querie
passage_embeddings = model.encode(input_passa

# calculate cosine similarity
print(util.cos_sim(query_embeddings, passage_
```

Usage with Huggingface Transformers: This is a simple example of how to use the Granite-Embedding-107m-Multilingual model with the Transformers library and PyTorch.

First, install the required libraries

```
pip install transformers torch
```

The model can then be used to encode pairs of text

```
import torch
from transformers import AutoModel, AutoToke

model_path = "ibm-granite/granite-embedding-1
```

maxpar1/leaderboard

joelg/discover_rag

kiedistv/leaderboard

malyvena/leaderboard

+5 Spaces

Collection including `ibm-granite/...`

Granite Embedding Collection
Embedding mo... • 9 items • U.. • 47

Paper for `ibm-granite/granite-e...`

Granite Embedding Models
Paper • 2502.20204 • Publis... • 5

Evaluation results

accuracy ...	test set	self-reported	72.714
f1 on MTE...	test set	self-reported	60.445
f1_weight...	test set	self-reported	77.854
ap on MTE...	test set	self-reported	22.496
ap_weight...	test set	self-reported	22.496
main_scor...	test set	self-reported	72.714
accuracy ...	test set	self-reported	71.672
f1 on MTE...	test set	self-reported	65.422

Expand 10455 evaluations

```
# Load the model and tokenizer
model = AutoModel.from_pretrained(model_path
tokenizer = AutoTokenizer.from_pretrained(mod
model.eval()

input_queries = [
    ' Who made the song My achy breaky heart
    'summit define'
    ]

# tokenize inputs
tokenized_queries = tokenizer(input_queries,

# encode queries
with torch.no_grad():
    # Queries
    model_output = model(**tokenized_queries
    # Perform pooling. granite-embedding-107
    query_embeddings = model_output[0][:, 0]

# normalize the embeddings
query_embeddings = torch.nn.functional.norma
```

Evaluation: The average performance of the Granite-Embedding-107M-Multilingual on Multilingual Miracl (across 18 langauges), Mintaka Retrieval (across 8 languages) and MTEB Retrieval for English (across 15 tasks), German (across 4 tasks), Spanish (across 2 tasks), Frenc (across 5 tasks), Japanese (across 2 tasks), Arabic (1 task), Korean (1 task) and Chinese (across 8 tasks) is reported below. Granite-Embedding-107M-Multilingual is twice as fast as other models with similar embedding dimensions.

Model	Paramters (M)	Embedding Dimension	Miracl (18)	Mintal Retrie (8)
granite-embedding-107m-multilingual	107	384	55.9	22.6

Model Architecture: Granite-Embedding-107m-Multilingual is based on an encoder-only XLM-RoBERTa like transformer architecture, trained internally at IBM Research.

Model	granite-embedding-30m-english	granite-embedding-125m-english	granite-embedding-107m-multilingua
Embedding size	384	768	**384**
Number of layers	6	12	**6**
Number of attention heads	12	12	**12**
Intermediate size	1536	3072	**1536**
Activation Function	GeLU	GeLU	**GeLU**
Vocabulary Size	50265	50265	**250002**
Max. Sequence Length	512	512	**512**
# Parameters	30M	125M	**107M**

Training Data: Overall, the training data consists of four key sources: (1) unsupervised title-body paired data scraped from the web, (2) publicly available paired with permissive, enterprise-friendly license, (3) IBM-internal paired data targetting specific technical domains, and (4) IBM-generated synthetic data. The data is listed below:

Dataset	Num. Pairs
Multilingual MC4	52,823,484
Multilingual Webhose	12,369,322
English Wikipedia	20,745,403
Multilingual Wikimedia	2,911,090
Miracl Corpus (Title-Body)	10,120,398
Stack Exchange Duplicate questions (titles)	304,525
Stack Exchange Duplicate questions (titles)	304,525

Dataset	Num. Pairs
Stack Exchange Duplicate questions (bodies)	250,519
Machine Translations of Stack Exchange Duplicate questions (titles)	187,195
Stack Exchange (Title, Answer) pairs	4,067,139
Stack Exchange (Title, Body) pairs	23,978,013
Stack Exchange (Title, Body) pairs	23,978,013
Machine Translations of Stack Exchange (Title+Body, Answer) pairs	1,827,15
SearchQA	582,261
S2ORC (Title, Abstract)	41,769,185
WikiAnswers Duplicate question pairs	77,427,422
CCNews	614,664
XSum	226,711
SimpleWiki	102,225
Machine Translated Cross Lingual Parallel Corpora	28,376,115
SPECTER citation triplets	684,100
Machine Translations of SPECTER citation triplets	4,104,600
Natural Questions (NQ)	100,231
SQuAD2.0	87,599
HotpotQA	85,000
Fever	109,810
PubMed	20,000,000
Multilingual Miracl Triples	81,409
Multilingual MrTydi Triples	48,715
Sadeeem Question Asnwering	4,037
DBPedia Title-Body Pairs	4,635,922
Synthetic: English Query-Wikipedia Passage	1,879,093
Synthetic: English Fact Verification	9,888
Synthetic: Multilingual Query-Wikipedia Passage	300,266
Synthetic: Multilingual News Summaries	37,489

Dataset	Num. Pairs
IBM Internal Triples	40,290
IBM Internal Title-Body Pairs	1,524,586

Notably, we do not use the popular MS-MARCO retrieval dataset in our training corpus due to its non-commercial license, while other open-source models train on this dataset due to its high quality.

Infrastructure: We train Granite Embedding Models using IBM's computing cluster, Cognitive Compute Cluster, which is outfitted with NVIDIA A100 80gb GPUs. This cluster provides a scalable and efficient infrastructure for training our models over multiple GPUs.

Ethical Considerations and Limitations: The data used to train the base language model was filtered to remove text containing hate, abuse, and profanity. Granite-Embedding-107m-Multilingual is finetuned on 12 languages, and has a context length of 512 tokens (longer texts will be truncated to this size).

Resources

- ⭐️ Learn about the latest updates with Granite: https://www.ibm.com/granite
- 📄 Get started with tutorials, best practices, and prompt engineering advice: https://www.ibm.com/granite/docs/
- 💡 Learn about the latest Granite learning resources: https://ibm.biz/granite-learning-resources